

SCIENCE REPORT

☐ OPEN

☒ U/16 - Collection only

(Please ✓ the appropriate box)

☐ Collection

☐ Engineering

☒ Research

☐ Theoretical

Name:	MANSHIKA NARAYAN		
Name:			
School:	S-C.C.S	Cust. Code:	SUV 01

State your purpose, hypothesis, or reason for your project:

To investigate how different filtering materials help clean dirty water. If dirty water passes through layers of gravel, sand, cotton, and charcoal, then the water will become cleaner because the materials trap dirt and particles.

Record the process or steps used in solving or approving the problem (hypothesis) or include an illustration of how your project works:

1. Cut the plastic bottle in half.
2. Turn the top half upside down like a funnel.
3. Place cotton or cloth at the bottom.
4. Add layers in this order:
 - o Cotton
 - o Charcoal
 - o Fine Sand
 - o Gravel
5. Pour dirty water slowly into the filter.
6. Observe the water collected at the bottom.
7. Compare the dirty water with the filtered water.

What has been proved or illustrated, or what is the conclusion drawn from your project:

The experiment showed that filtration can help clean dirty water. Sand, gravel, charcoal, and cotton work together to remove dirt and impurities. This demonstrates how water treatment systems help provide cleaner water for people.

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SCIENCE REPORT - Continued

Give a Scriptural application or reference for your project:
Clean water is a blessing from God and is important for health and life.
Matthew 10:42 "Whoever gives even a cup of cold water....."

Give a brief history of the discovery/invention or the hypothesis/facts you are using in your project. Describe how the discovery/invention advanced to today's use, give your opinion of its future use:
Water filtration began in ancient times when people used sand, gravel and cloth to clean water. The Greek physician Hippocrates (460-370BC) created a cloth filter called the "Hippocratic sleeve" to remove impurities. In the 1800s, slow sand filtration was used in Europe to reduce diseases like cholera. Today, modern systems use multiple stages such as sand, gravel, activated carbon and chemical disinfection to produce safe drinking water.
This project is based on the idea that different materials trap different particles. Gravel removes large dirt, sand removes fine particles, and charcoal absorbs impurities. Water filtration remains essential and will improve with future technology.

Manshika			
(Student's Signature)	(Date)	(Student's Signature)	(Date)

* Note: A written paper may be submitted as a substitute for this science report. The paper should at least cover the above information.